

Github Link:

<https://github.com/eguitar/cs580f25quiz6>

Part #1

```
package quiz6;

import java.util.ArrayList;

public class Stock {
    private String symbol;
    private double price;
    private ArrayList<Observer> observers;

    public Stock(String symbol, double price) {
        this.symbol = symbol;
        this.price = price;
        this.observers = new ArrayList<>();
    }

    public void addTrader(Observer observer) { observers.add(observer); }

    public void setPrice(double price) {
        if (this.price != price) {
            this.price = price;
            notifyObservers();
        }
    }

    public double getPrice() { return price; }

    private void notifyObservers() {
        for (Observer observer : observers) {
            observer.update(symbol, price);
        }
    }
}
```

```
package quiz6;

public interface Observer {
    void update(String symbol, double price);
}
```

```
package quiz6;

public class StockTrader implements Observer {
    private String name;
    private String symbol;
```

```
private double price;

public StockTrader(String name, String symbol) {
    this.name = name;
    this.symbol = symbol;
    this.price = -1;
}

@Override
public void update(String symbol, double price) {
    String output = "Stock Trader Name: " + this.name + "\n" +
        "Stock Observing: " + this.symbol + "\n" +
        "Current Price: " + price + "\n";

    if (this.price == -1) { output += "Initial price set"; }
    else if (price > this.price) { output += "The price went up."; }
    else if (price < this.price) { output += "The price went down."; }
    else { output += "The price stayed the same."; }

    System.out.println(output);
    this.price = price;
}
}
```

```
package quiz6;

import java.util.Random;

public class Main {
    public static void main(String[] args) {
        Stock stock = new Stock("AMD", 203.78);

        StockTrader trader1 = new StockTrader("Zach Tran", "AMD");
        StockTrader trader2 = new StockTrader("Viet Nguyen", "AMD");
        StockTrader trader3 = new StockTrader("Johnathan Ko", "AMD");

        stock.addTrader(trader1);
        stock.addTrader(trader2);
        stock.addTrader(trader3);

        Random random = new Random();
        double currentPrice = stock.getPrice();

        for (int i = 0; i < 3; i++) {
            double change = (random.nextDouble() * 20) - 10;
            System.out.println("-----");
            System.out.print("Old Price: " + currentPrice + " ----- New Price: ");
        }
    }
}
```

```
        currentPrice = Math.round((currentPrice + change) * 100.0) / 100.0;
        System.out.println(currentPrice);
        System.out.println("-----");
        stock.setPrice(currentPrice);

        try {
            Thread.sleep(1000);
        } catch (InterruptedException e) {
            Thread.currentThread().interrupt();
        }
    }
}
```

Output”

```
(base) eguitar@MacBookPro cs580 % mvn exec:java -Dexec.mainClass="quiz6.Main"
[INFO] --- exec:3.6.2:java (default-cli) @ cs580 ---
-----
Old Price: 203.78 ----- New Price: 194.48
-----
Stock Trader Name: Zach Tran
Stock Observing: AMD
Current Price: 194.48
Initial price set
Stock Trader Name: Viet Nguyen
Stock Observing: AMD
Current Price: 194.48
Initial price set
Stock Trader Name: Johnathan Ko
Stock Observing: AMD
Current Price: 194.48
Initial price set
-----
Old Price: 194.48 ----- New Price: 191.6
-----
Stock Trader Name: Zach Tran
Stock Observing: AMD
Current Price: 191.6
The price went down.
Stock Trader Name: Viet Nguyen
Stock Observing: AMD
Current Price: 191.6
The price went down.
Stock Trader Name: Johnathan Ko
Stock Observing: AMD
Current Price: 191.6
The price went down.
-----
Old Price: 191.6 ----- New Price: 187.19
-----
Stock Trader Name: Zach Tran
Stock Observing: AMD
Current Price: 187.19
The price went down.
Stock Trader Name: Viet Nguyen
Stock Observing: AMD
Current Price: 187.19
The price went down.
Stock Trader Name: Johnathan Ko
Stock Observing: AMD
Current Price: 187.19
The price went down.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.248 s
```